CHAPTER 7

Voice, Cameras, and the Smart-Home Lesson

What doorbells and smart speakers taught us about trust, and how to design voice and camera features that save money and energy from the first sketch

In this chapter

- Two smart-home cases that were design failures, not hacks
- How attackers talk to voice assistants without making a sound
- Who is really in the aisle: designing for six kinds of people
- Where the audio goes: a privacy-first voice design for Glidecart
- Ten design requirements, each with the money or energy it saves
- How to test voice features with people before building hardware

Millions of households now talk to their homes. We ask a speaker for the weather, check a doorbell camera from work, and tell a phone to add milk to the list. Voice and video have become the most natural way to use technology, especially for people whose hands are full, whose eyes are tired, or who never learned to type quickly.

Stores want the same convenience. A cart you can talk to (“Where is the almond milk?”) or a deli kiosk that takes your order by voice could make shopping easier for everyone. But a cart with a microphone and a camera, rolling through a public store full of families, carries every risk of a smart-home device and a few more. This chapter shows how to keep the convenience and design out the risk, using two real cases from the smart-home industry as a guide.

Two Cases That Were Design Failures, Not Hacks

The doorbell camera

In May 2023, the Federal Trade Commission (FTC) charged that Ring, the doorbell and home-camera company, had given employees and contractors broad access to customers' private videos, including videos from inside homes, and had failed to put basic protections in place against attackers who took over customer accounts and cameras. Ring agreed to pay USD 5.8 million, which the FTC said would be used for customer refunds, and to strengthen its privacy and security program (Federal Trade Commission [FTC], 2023a).

Notice what the case was about. It was not a clever new exploit. It was about **who inside the company could see what**, and whether customers' accounts were protected against password guessing and reuse. Both were design decisions. Both could have been settled in a planning meeting.

The smart speaker

The same week, the FTC and the Department of Justice charged that Amazon had kept children's Alexa voice recordings indefinitely and had undermined parents' requests to delete them. Amazon agreed to a USD 25 million civil penalty and to delete certain data (FTC, 2023b).

Again, the core issue was a design choice: **how long to keep recordings, and whether "delete" really means delete.** Keeping everything forever felt free at the time. It was not.

The lesson for any product with a microphone or camera

Most of the damage in these cases came from decisions about **access** (who can see or hear the data), **retention** (how long it is kept), and **account protection** (MFA and defenses against password reuse). Each is cheapest to get right before the first device ships.

There is a further lesson for threat intelligence. When I searched the CISA catalog of actively exploited vulnerabilities for smart-home and voice ecosystems, it had **no entries** at all, while routers and wireless gateways had 26 entries added in the last two years (Chapter 4). The biggest smart-home failures are not in vulnerability lists. They are in regulators' case files, and designers should read those too.

Talking to a Machine Without Making a Sound

Voice assistants have a property that keyboards do not: they accept commands from anyone within earshot, and sometimes from things that are not people at all.

In 2017, researchers showed that they could send commands to popular voice assistants using ultrasound, sound above the range of human hearing. The assistants heard and obeyed commands that nobody in the room could hear. The researchers named it DolphinAttack (Zhang et al., 2017). In 2020, another team showed they could inject commands by aiming a modulated laser at a device's microphone, in some cases from more than 100 meters away and through a window (Sugawara et al., 2020).

These attacks need equipment and effort, so they are not the most likely threat to a grocery cart. But they teach a design principle that costs nothing: **never let voice alone do anything that would be harmful if the wrong person said it.** Finding the almond milk is harmless. Paying for a basket, unlocking a cart, or changing settings is not. If those actions require a tap on the screen or a staff badge, the attacks lose their value, and no special anti-spoofing hardware is needed.

Who Is Really in the Aisle?

The first stage of design thinking is to empathize: to understand the people who will use, maintain, and live around a product (Brown, 2008). ISO 9241-210 makes the same point for interactive systems: design from an explicit understanding of users, tasks, and environments, and involve users throughout (International Organization for Standardization [ISO], 2019). For Glidecart, that means at least six kinds of people, shown in Figure 7.1.

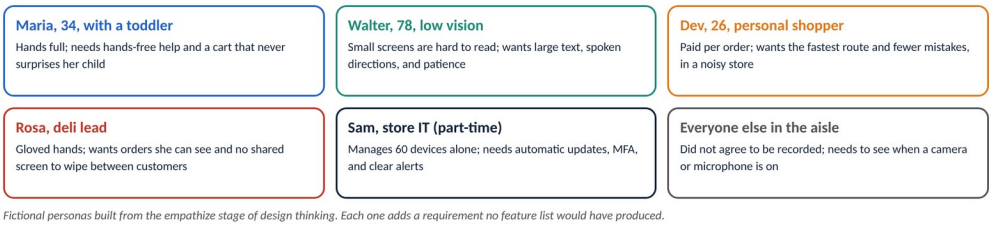


Figure 7.1 Six people in the aisle. Each persona is fictional, but each adds a requirement: hands-free help (Maria), large text and spoken directions (Walter), speed in noise (Dev), touchless ordering (Rosa), automatic updates and MFA (Sam), and visible recording indicators (everyone else).

Two people on that list deserve special attention. **Walter** shows why voice is an accessibility feature, not a gimmick: for many people with low vision or limited hand movement, speaking is easier than tapping a small screen. Accessibility guidelines such as the Web Content Accessibility Guidelines encourage multiple ways to perceive and operate an interface (World Wide Web Consortium [W3C], 2023). **Everyone else in the aisle** shows why privacy is not only about the person using the cart. Shoppers walking past did not agree to anything. They deserve to know when a camera or microphone is active, and to be confident that nothing about them is stored.

Voice does not work equally well for everyone

A study of five commercial speech-recognition systems found that they made nearly twice as many errors for Black speakers as for white speakers, with an average word error rate of 0.35 compared with 0.19 (Koenecke et al., 2020). Accents, age, background noise, and speech differences all affect accuracy. A voice feature that works well in a quiet lab may frustrate exactly the shoppers it was meant to help. The design response is to **always offer a second way**: a large-text screen, a push-to-talk button, and a staff call button, and to test with a diverse group of real shoppers before launch (Study S1, later in this chapter).

Where Does the Audio Go?

The single most important design decision for a voice product is where speech is processed. In many consumer devices, audio is sent to a company's servers to be interpreted. That is convenient for the company, but it creates a stream of recordings that must be secured,

retained, and eventually deleted, which is exactly what went wrong in the Alexa case.

Glidecart's design keeps audio on the cart. Figure 7.2 shows the flow.

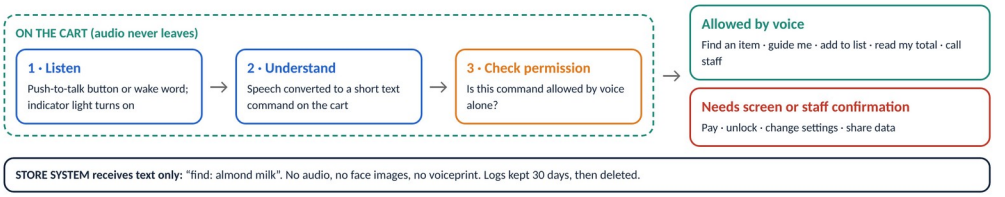


Figure 7.2 Privacy-first voice flow for Glidecart. Speech is converted to a short text command on the cart; only the text command reaches the store system, and only voice-safe commands run without confirmation.

This design does three things at once. It **protects privacy**, because there are no recordings to leak, misuse, or subpoena. It **saves money**, because the store does not pay for cloud speech services or for a large privacy program to manage audio archives. And it **saves energy**, because the cart's radio is not streaming audio all day; it sends a few bytes of text when a command is made. On-device speech recognition for a small vocabulary of shopping commands is well within the ability of current embedded processors.

> **Design tip: the indicator light is a feature**
>
> A small light that turns on whenever the microphone or camera is active costs cents. It tells shoppers and bystanders what is happening without words, in any language. Make it hard-wired to the sensor's power, so software cannot turn the sensor on while the light is off.

Cameras: See the Product, Not the Person

Glidecart uses a camera to recognize items placed in the basket and a LiDAR sensor to avoid people and displays. Neither job requires recognizing faces. The design therefore follows a simple rule: **point cameras at products, process images on the cart, and store no images of people.** When the cart is parked in the corral, cameras turn off. When it is moving, the vision system runs only when something enters or leaves the basket, rather than recording continuously.

That rule also removes a legal risk. Some laws strictly regulate biometric identifiers such as face geometry and voiceprints. Illinois, for example, requires written consent before collecting them and allows individuals to sue over violations (Biometric Information Privacy Act, 2008). A cart that never creates biometric data never has to manage that consent. Chapter 3 described how an earlier concept to detect shoppers' emotional “impulse” states with biometric sensors was removed for the same reason.

Cameras can also be fooled. Researchers have shown that small, carefully designed stickers can make vision systems misread objects (Eykholt et al., 2018). For a shopping cart, the practical defense is a second check: the cart's weight sensor confirms that what the camera saw matches what was placed in the basket, and mismatches are flagged for a quick review at checkout.

Ten Design Requirements That Save Money and Energy

Table 7.1 turns the lessons of this chapter into requirements an engineer can build and an auditor can check. The last column shows why each is cheaper to do now than later.

Table 7.1 Voice and camera design requirements for Glidecart and the deli kiosk

#	Requirement	Lesson or source	Money or energy saved from the start
1	Speech and wake word processed on the device; no audio uploaded	Alexa case (FTC, 2023b)	No cloud speech fees; no audio archive to secure; less radio energy
2	Push-to-talk button on the handle, plus wake word	Noisy stores; accidental activation	Fewer false wake-ups and wasted processing
3	Voice cannot pay, unlock, change settings, or share data	DolphinAttack; Light Commands	No anti-spoofing hardware needed
4	Staff and admin accounts are individual and use MFA; access is logged	Ring case (FTC, 2023a)	Avoids breach, refund, and regulatory costs
5	Hard-wired indicator light whenever microphone or camera is on	Bystander trust	Cents per unit; fewer opt-outs that waste the investment
6	Cameras face the basket; no stored images of people; no biometrics	Biometric privacy laws	No consent program; no biometric data to protect
7	Weight sensor double-checks what the camera sees	Adversarial stickers	Fewer checkout disputes; no extra cameras
8	Large text, spoken prompts, multiple	Accessibility; speech-recognition gaps	One design serves more shoppers; avoids redesign

	languages, staff call button		
9	Text logs kept 30 days, then deleted automatically	Retention lessons from both cases	Less storage; less to disclose or defend
10	Sensors off in the corral; event-triggered vision when moving	Energy-aware design	Smaller battery; lower charging energy

Notice that none of these requirements needs exotic technology. Most remove something: a recording, a permission, a data type, an always-on sensor. Removing things early is the cheapest form of security there is, and it is often the cheapest form of energy saving too.

At the Deli Counter: Touchless Ordering

The same thinking applies to the deli ordering kiosk introduced in Chapter 3. Deli staff wear gloves and handle food; customers touch shared screens between handling money and phones. The FDA Food Code sets cleaning expectations for food-contact surfaces and the food-handling area (U.S. Food and Drug Administration, 2022). A design that lets customers order by voice, by scanning a code with their own phone, or by gesture reduces how many people touch the shared screen, and the screen itself can be cleaned on the store's normal schedule.

Whether touchless ordering actually shortens lines and improves satisfaction is a hypothesis, not a fact. Chapter 11 describes a field test to find out. The privacy rules from this chapter still apply: voice orders are processed locally, the kiosk keeps the order text but not the audio, and nothing identifies the customer unless they choose to sign in.

Test It With People Before You Build It

All of these requirements can be tested before a single cart is built. A **Wizard-of-Oz** prototype is a fake product that looks real to the participant while a researcher quietly produces its responses (Dahlbäck et al., 1993). For Glidecart, a tablet on an ordinary shopping cart can show the interface and speak directions while a researcher out of sight triggers each response.

The planned study, Study S1, asks participants to find five items in a mock aisle twice, once with touch only and once with voice plus touch, in a random order. It measures task time, errors, and the System Usability Scale, a ten-question survey widely used to compare designs (Brooke, 1996). To detect a medium-sized difference with standard statistical confidence, the study needs about **34 participants**. The sample should deliberately include older adults, people with low vision or limited hand movement, and speakers with a range of accents. The cost is a tablet, a cart, and gift cards for participants, plus approval from an ethics review board.

If you do only three things

1. Keep audio and images on the device; send only what the store needs, as text.
2. Do not let voice alone pay, unlock, or change anything.
3. Test with the people least like your design team before you build.

Your Turn: Voice and Camera Design Review

Use these questions in a design review for any product with a microphone or camera.

Question	Good answer
Where is audio or video processed?	On the device; only text or counts leave it
What is stored, and for how long?	The minimum, with automatic deletion on a stated schedule
Who inside the company can see or hear it?	Named roles only, with MFA and access logs
What can voice do without confirmation?	Only harmless actions
How do bystanders know it is on?	A hard-wired indicator light and plain signage
Does it create biometric data?	No, or only with explicit opt-in consent
What is the second way to use it?	Screen, button, or staff help for anyone voice fails
When is it off?	Whenever it is not being used

Chapter 7 Checklist

- ☐ Audio and video are processed on the device by default.
- ☐ Voice cannot complete payments, unlock devices, or change settings on its own.
- ☐ Staff and administrator accounts are individual, use MFA, and are logged.
- ☐ A hard-wired indicator shows when a microphone or camera is active.

☐ No images of people or biometric data are stored without explicit consent.

☐ Retention periods are written down and deletion is automatic.

☐ There is always a second way to use every voice feature.

☐ Sensors switch off when the device is idle.

☐ The design has been tested with a diverse group of real users.

Try It

Exercise 1 (20 minutes). Open the privacy settings of a voice assistant or doorbell camera you or your family use. Find out where recordings are stored, how long they are kept, and how to delete them. How many taps did it take? What would you change?

Exercise 2 (45 minutes). Build a paper Wizard-of-Oz test. Tape a printed "screen" to a shopping basket, ask a friend to find three items in your kitchen using only spoken directions from you, then only the printed screen. Time each task and ask which they preferred, and why.

Questions for Discussion

- The Ring and Alexa cases were about access and retention rather than hacking. Why are these decisions so often left until late in a project?
- Should a voice assistant ever be allowed to complete a payment? Under what conditions?
- How would you explain the indicator light to a shopper who does not speak English?
- Speech recognition works better for some groups than others. Who is responsible for fixing that: the vendor, the store, or the designer?

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Note on the cases: Glidecart and the deli kiosk are unbuilt concepts by the author. Personas are fictional. Ring and Alexa facts are summarized from FTC announcements; readers should consult the original orders for full terms.